UNIFORM VANISHING OF A TOR MAP IN THE DEGREE OF AN UPPER BOUND FOR THE NUMBER OF GENERATORS OF THE IDEAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-local-cohomology/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-local-cohomology/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

Editorial note. Bibliography provenance: neither the counted statement nor its proof contains a citation (this lane rescans the whole span before writing the card), so the card carries no bibliography and no entry was reconstructed.

Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The annihilation of the Tor modules above the number of generators of the ideal by a fixed power of the ideal, uniformly in the module and the exponent. Author lemma lemma-annihilates-tors, whose statement is reproduced verbatim as a context span of this card (source lines 4668-4675). Supplies the single constant c_1 used in the definition of the invariant of the module and in the diagram of the induction step.
- The vanishing of the Tor maps for a module annihilated by a fixed power of the ideal, again uniformly in the module. Author lemma lemma-maps-zero-fixed-torsion, whose statement is reproduced verbatim as a context span of this card (source lines 4510-4519). Supplies the constants that give the base case of the induction.

Lemma 0.1. *Let I be an ideal of a Noetherian ring A. Let $t \geq 0$ be an upper bound on the number of generators of I. There exist $N, c \geq 0$ such that the maps*

$$Tor_{t+1}^A(M, A/I^n) \to Tor_{t+1}^A(M, A/I^{n-c})$$

are zero for any A-module M and all $n \geq N$.

Lemma 0.2. *In the situation above, let t be an upper bound on the number of generators for I. There exists an integer $c = c(A, I) \geq 0$ such that for any A-module M the tor modules $Tor_i^A(M, A/I^q)$ are annihilated by I^c for $i > t$ and all $q \geq 0$.*

Date: stacks-project:local-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.3. *Let I be an ideal of a Noetherian ring A. For every $m \geq 0$ and $i > 0$ there exist a $c = c(A, I, m, i) \geq 0$ such that for every A-module M annihilated by I^m the map*

$$Tor_i^A(M, A/I^n) \to Tor_i^A(M, A/I^{n-c})$$

is zero for all $n \geq c$.

Proof. Let c_1 be the constant found in Lemma 0.2. Please keep in mind that this constant c_1 works for Tor_i for all $i > t$ simultaneously.

Say $I = (a_1, \ldots, a_t)$. For an A-module M we set

$$\ell(M) = \#\{i \mid 1 \leq i \leq t,\ a_i^{c_1} \text{ is zero on } M\}$$

This is an element of $\{0, 1, \ldots, t\}$. We will prove by descending induction on $0 \leq s \leq t$ the following statement H_s: there exist $N, c \geq 0$ such that for every module M with $\ell(M) \geq s$ the maps

$$\mathrm{Tor}_{t+1+i}^A(M, A/I^n) \to \mathrm{Tor}_{t+1+i}^A(M, A/I^{n-c})$$

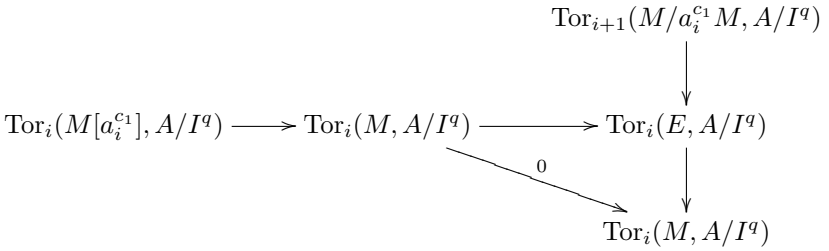
are zero for $i = 0, \ldots, s$ for all $n \geq N$.

Base case: $s = t$. If $\ell(M) = t$, then M is annihilated by $(a_1^{c_1}, \ldots, a_t^{c_1}\}$ and hence by $I^{t(c_1-1)+1}$. We conclude from Lemma 0.3 that H_t holds by taking $c = N$ to be the maximum of the integers $c(A, I, t(c_1 - 1) + 1, t + 1), \ldots, c(A, I, t(c_1 - 1) + 1, 2t + 1)$ found in the lemma.

Induction step. Say $0 \leq s < t$ we have N, c as in H_{s+1}. Consider a module M with $\ell(M) = s$. Then we can choose an i such that $a_i^{c_1}$ is nonzero on M. It follows that $\ell(M[a_i^c]) \geq s + 1$ and $\ell(M/a_i^{c_1}M) \geq s + 1$ and the induction hypothesis applies to them. Consider the exact sequence

$$0 \to M[a_i^{c_1}] \to M \xrightarrow{a_i^{c_1}} M \to M/a_i^{c_1}M \to 0$$

Denote $E \subset M$ the image of the middle arrow. Consider the corresponding diagram of Tor modules



with exact rows and columns (for every q). The south-east arrow is zero by our choice of c_1. We conclude that the module $\mathrm{Tor}_i(M, A/I^q)$ is sandwiched between a quotient module of $\mathrm{Tor}_i(M[a_i^{c_1}], A/I^q)$ and a submodule of $\mathrm{Tor}_{i+1}(M/a_i^{c_1}M, A/I^q)$. Hence we conclude H_s holds with N replaced by $N + c$ and c replaced by $2c$. Some details omitted. □